Response Summary:

Research Ethics Application Form - CEP

Principle Investigator and Student Details
(to be completed by all students)

Section 1a: Details of Principal Investigator (PI)

For all projects the PI must be employed by Imperial College London or hold an honorary contract.
For all student projects, the student's supervisor must be named as PI.
Student, co-investigator and collaborator details must be added to Section 13.

Q2. Supervisor (PI) name (incl. title)
Dr Iain Staffell

Q3. Academic position (at Imperial College London)
Associate PRofessor in Sustainable Energy

Q4. Faculty
Faculty of Natural Sciences

Q5. Division/ School/ Department
Centre for Environmental Policy

Q6. PI Email (Imperial College)
i.staffell@imperial.ac.uk

Q7. Summary of PI skills (experience relevant to the study and in any procedures to be used) (350 characters max)
Iain Staffell is a multi-disciplinary scientist holding degrees in Physics, Chemical Engineering and Economics. He is an Associate Professor of Sustainable Energy at the Centre for Environmental Policy with fifteen years' experience in energy R&D.

Section 1b: Student Details

Q253. Student name
Yunxi Shen

Q254. Student email (Imperial College)
ys4625@ic.ac.uk

Section 2: Research Type

Q9. Is your study considered low risk?
'Low Risk' includes collecting data from individuals or groups that can give consent, or using secondary data that you have permission to use.
See the Research Term Guidance document for more information and consult with your supervisor.

- Yes

Q10. Does your study only involve analysis of secondary data which is publicly available, and permission is not required to access the data? This includes extended or systematic literature reviews.

- Yes

Section 3: Filter for ICREC and SETREC

Q13. Is the primary aim of the research answering a human health related question?

- No

Q14. Is the primary aim of the research answering a non-health related science, social science, engineering or technology related question?

- Yes

Q15. Is the primary aim of the research to answer an educational question?

- No

Section 4: Risk level categorisation

This section determines the research risk level and if the application requires full committee review.

Meeting dates and submission deadlines ICREC/SETREC.

Q18. High risk

Does the research involve drugs/medication? If yes, please attach the SmPc.	No
Does the research involve genetically modified materials? If yes, please also complete appendix two and attach the GM Safety Committee letter.	No
Will you be recruiting vulnerable participants? i.e. children (15 years or younger), adults (16 years or over) who are unable to consent, people in care, the mentally ill or individuals with learning difficulties?	No
Will participants take part in the study without their explicit consent? i.e. studies involving deception.	No
Will you be recruiting prisoners or young offenders?	No
Is there any aspect of the proposed research which could potentially cause harm to the reputation of the College? i.e. could the research be considered controversial or prejudiced?	No
Could participants disclose any illegal or harmful activity due to the nature of the research?	No
Will personally sensitive subjects be discussed that have the potential to induce stress, anxiety or negative consequences for the participant?	No

Will the researcher be in a position of influence or authority over the participants that could give rise to a perceived pressure to participate? i.e. lecturers/teachers and students.	No
Does the study involve physically intrusive procedures, administration of substances, use of bodily fluids, tissues, DNA or RNA? Use of relevant material must be registered with Imperial College Tissue Bank under the College HTA license.	No
Does the study involve ultrasound or sources of non-ionizing radiation? i.e. radiation, MRI, or fMRI.	No
Are there any potential conflicts of interest, or what could be perceived by an outside observer as conflicts of interest?	No
Will undue incentives for participants be offered? Incentives should be proportionate to the burden imposed and justified by the benefits.	No
Are you using any medical device in the UK that is CE/UKCA marked but is being used outside its product limitation? Or are you using any non-CE/non-UKCA marked product(s)? For more information on regulating medical devices.	No
Does the proposed research raise any ethical issues that are not covered above?	No

If you have answered <u>NO TO ALL</u> of the above questions, your project is considered low risk and you can proceed with the rest of the ethics form.

Section 5: Project Description

(to be completed by all)

Q22. Full title of study
Climate Mitigation Wedges for Singapore's Net Zero Transition

Q23. P code or cost code and study funder
(only if applicable to study)
N/A

Q24. Lead organisation
(who has overall responsibility for the study)
Imperial College London

Q25. List of location(s) where study will be conducted
London, United Kingdom

Q26. Proposed start date
From start of advertising and/or recruitment
Immediate

Q27. Proposed end date
To end of data collection
2 September 2026

Section 6: Project Summary

Q29. Provide a summary of the project in lay terms, this should include:

- **a brief description of reasons for doing the study (justification for the research)**
- **the aims (research aim/question)**
- **how data will be disseminated**
- **any expected benefits to the participant, researchers or others.**

(500 words max)

Singapore’s latest GHG inventory in 2023 showed that its total emissions was 55.5 MtCO2e. (Department of Statistics Singapore, 2025) In Singapore’s 2nd NDC submitted on 10 Feb 2025, it has committed to reduce absolute emissions to ~60MtCO2e in 2030 after peaking earlier, then to ~45-50MtCO2e in 2035, and finally achieve net zero emissions in 2050. (National Climate Change Secretariat Singapore, 2025) While its net zero roadmap contain many high-level targets, policies and plans in various sectors, for example 6 GW of clean energy electricity imports by 2035 and ~2Mt of cross-border CCUS by 2030, they lack granular details on the anticipated pathways and extent of implementation, impacting its credibility. (Climate Action Tracker, 2024)

What are the mitigation strategies, in terms of policies and technologies, that Singapore can deploy to achieve its target of net zero by 2050? The climate stabilisation wedges framework offers a simple language for global decarbonisation by breaking the challenge into clear strategies, each yielding 2 GtCO2e/year abatement by 2050. (Johnson & Staffell, 2026) In exploring the research question, the project uses the climate stabilisation wedges framework (on a global scale) to apply it to Singapore on a national scale. It considers the technical, economic, and political feasibility of each wedge, within Singapore’s social and regulatory context. The outline of key steps in the project are as follows:

Rescale a wedge of mitigation from global to national level. Preliminarily, some contenders for the principle(s) to do this are population size, GDP, and land area.

Estimate how many wedges are needed for the country to get from current policy to net zero.

Reassess the effort needed for each strategy to achieve a wedge, and the upper bound on their deployment, within that country’s context.

Through the above, identify what Singapore’s decarbonisation ‘menu’ could potentially look like to close the gap between current policy to net zero with more certainty or concrete estimates, and highlight what the key sensitivities, trade-offs in resources required, feasibility and potential financing implications unusually promising strategies could be.

Singapore, lacking the land for many strategies like land-based ecosystem restoration and renewable energy (wind, geothermal, tidal), is a critical case study with unique, severe decarbonisation challenges. In doing this project, it can also provide actionable, diversified, portfolio-based recommendations for Singapore's net zero transition, including how to allocate climate financing. More broadly, this may possibly yield lessons for other small island developing states (SIDS), which comprise 30% of the world’s countries, and deal with similar challenges as Singapore.

Singapore’s latest GHG inventory in 2023 showed that its total emissions was 55.5 MtCO2e. (Department of Statistics Singapore, 2025) In Singapore’s 2nd NDC submitted on 10 Feb 2025, it has committed to reduce absolute emissions to ~60MtCO2e in 2030 after peaking earlier, then to ~45-50MtCO2e in 2035, and finally achieve net zero emissions in 2050. (National Climate Change Secretariat Singapore, 2025) While its net zero roadmap contain many high-level targets, policies and plans in various sectors, for example 6 GW of clean energy electricity imports by 2035 and ~2Mt of cross-border CCUS by 2030, they lack granular details on the anticipated pathways and extent of implementation, impacting its credibility. (Climate Action Tracker, 2024)

What are the mitigation strategies, in terms of policies and technologies, that Singapore can deploy to achieve its target of net zero by 2050? The climate stabilisation wedges framework offers a simple language for global decarbonisation by breaking the challenge into clear strategies, each yielding 2 GtCO2e/year abatement by 2050. (Johnson & Staffell, 2026) In exploring the research question, the project uses the climate stabilisation wedges framework (on a global scale) to apply it to Singapore on a national scale. It considers the technical, economic, and political feasibility of each wedge, within Singapore’s social and regulatory context. The outline of key steps in the project are as follows:

Rescale a wedge of mitigation from global to national level. Preliminarily, some contenders for the principle(s) to do this are population size, GDP, and land area.

Estimate how many wedges are needed for the country to get from current policy to net zero.

Reassess the effort needed for each strategy to achieve a wedge, and the upper bound on their deployment, within that country’s context.

Through the above, identify what Singapore’s decarbonisation ‘menu’ could potentially look like to close the gap between current policy to net zero with more certainty or concrete estimates, and highlight the key trade-offs in resources required, feasibility and potential financing implications.

Singapore, lacking the land for many strategies like land-based ecosystem restoration and renewable energy (wind, geothermal, tidal), is a critical case study with unique, severe decarbonisation challenges. In doing this project, it can also provide actionable, diversified, portfolio-based recommendations for Singapore's net zero transition, including how to allocate climate financing. More broadly, this may possibly yield lessons for other small island developing states (SIDS), which comprise 30% of the world’s countries, and deal with similar challenges as Singapore.

Section 7: Research Methods

Q31. What methods will you be using in this study?
Briefly describe in lay terms:

- **what will happen**
- **the number of times**
- **any data collection techniques**

(500 words max)

This project will use a primarily quantitative methodology based on modelling and secondary data analysis. The process will directly address the four stated objectives. Initially, a thorough literature review will be conducted to identify a suitable scaling principle (e.g., proportional to population, GDP, or land area) to define a 'Singapore-sized wedge' of mitigation.

The subsequent analysis involves four main steps:

Data Collection: Collect and collate key national-level data for Singapore, including current and projected greenhouse gas emissions, net zero targets, economic indicators, and detailed land-use constraints. Data will be sourced from official Singapore government reports (e.g., NEA, EDB) and international bodies (e.g., IEA, World Bank).

Total Wedges Calculation: Using the rescale factor and Singapore's net zero pathway, estimate the total number of national wedges required to close the gap between current policy and the net zero target.

Wedge Reassessment: Re-evaluate the technical and physical upper bound on deployment for a selection of core mitigation strategies (e.g., solar, CCUS, electric vehicles, energy efficiency) specifically within the context of Singapore's limited resources and geographical limitations. This involves simple quantitative calculations based on physical limits (e.g., available roof space for solar).

Synthesis: Combine the results to construct the final 'decarbonisation menu,' identifying which strategies are unusually promising and where the greatest sensitivities and trade-offs lie.

The full list of possible datasets required and their possible sources are at https://imperiallondon-my.sharepoint.com/:x:/g/personal/ys4625_ic_ac_uk/IQDMtO6m4YTaTpmZfJUtn6GYAfum49sMhW83TUUWwIm_Fq8?e=BzKhfz.

Q32.

STUDY OBJECTIVES

What are you hoping the study achieves? List the primary, secondary and other study objectives.

List the specific research objectives. Again, this can be copied from your research proposal

The primary aim of the study is to identify what Singapore's decarbonisation 'menu' could potentially look like, in order to close the gap between current policy to net zero with more certainty or concrete estimates. This would consider the feasibility of various mitigation strategies, including financing plans and the key tradeoffs involved.

Q33.

PUBLICATION POLICY

The study publication policy should be described in full.

In most cases, you should just state that this work is being prepared for publication as an MSc thesis at Imperial College London. If you intend to write a report for some external party or you intend to publish in the academic/peer reviewed literature you should say that too

This work is being prepared for publication as an MSc thesis at Imperial College London, and subsequent publication in academic/peer reviewed literature if suitable.

Q34.

STUDY DESIGN

Detail how this study will be conducted from advertising to data processing. Make sure your design is explicitly described and appropriately justified. Repeat for each method you use. Include the following bullet points below:
• Type of study method: survey, interviews, focus groups, simulator study.
• Recruitment methods: how you will recruit participants, use of participant information sheets and consent forms
• Duration of study from advertising and recruitment to end of data collection
• Number of participants. This should be justified.

This should describe what variables you will be measuring, how you will be measuring them, how often they will be measured (just once, or is there a follow-up or repeat measurement) and whether this will be done through survey, interview, focus group or other means. Describe here how many people you are looking to recruit and how you will recruit them. Any further details about inclusion criteria will be given in the next sections.

No participant recruitment involved. See question 1 of Section 7 re methodology.

Section 8: Participant Recruitment

Q36. Provide details of methods of recruitment, participant inclusion and exclusion criteria and the number of participants you are aiming to recruit.
Please include:

- **Details of any incentives (such as financial reimbursement).**
- **Recruitment and advertising material (email, poster, social media advert)**
- **Oral information scripts**

***The recruitment material (points 2 and 3) needs to be developed in a separate document and sent to your supervisor along with the print out of your ethics form.**
(1000 words max)

Not applicable

Q37.

Pre-recruitment evaluations

Are there any requirements that a participant must fulfil? E.g., fitness test, allergy test.

This will likely be not applicable to most of you – you should just say that there will be no pre-recruitment evaluations

Not applicable

Q38.

Inclusion criteria

Include justifications.

This is relevant if you are, for instance, interviewing professionals or members of a particular group. Indicate why you need to speak with these people specifically and how you will know that they meet your specific criteria. For instance, "I will contact the sustainability officers of large multinational companies based on their publicly available LinkedIn profiles. The study requires insights from professionals on [whatever the topic is…]"

Not applicable

Q39.

Exclusion Criteria

Include justifications.

If there is a reason why you might exclude someone after you contacted them, you should say this here. Otherwise, you may just reiterate that anyone meeting the inclusion criteria will be considered and there are no specific exclusion criteria. Extending the above example you might be looking for experts, so one of your exclusion criteria might be that if you have identified someone through LinkedIn as a sustainability officer, but it turns out they have less than 1 year experience in the role, you will exclude them

Not applicable

Section 9: Informed Consent

Include details of how you will be obtaining consent.

Q42. Detail the process for ensuring informed consent of all research participants.
Not applicable

Q43. The withdrawal process(es).
Describe procedures for
• stopping early
• when can a participant withdraw from the study
• how and who to contact.
Include details of data usage once participant has withdrawn if the data be used once it has been anonymised.
Not applicable

Q44. If participants whose first language is not English are to be recruited, state clearly how the details of the study will be explained, and the consent processed.
Not applicable

Section 10: Ethical Summary

Q46. Has any part of this proposal received prior ethics approval?
e.g. if project part of a larger project that has already gone through ethics, then an amendment to that ethics approval will need to be sought via College committees – may be lengthy – avoid if possible, unless minor.

- No

Q47. Is this study subject to local ethics approval to where research is being conducted?
e.g. through another university or institution

- No

Section 11: Documentation checklist

Q53. Do Imperial College's insurers need to be notified about your project?

If your project is running abroad and is not qualitative or data only, or if your project is interventional and involves pregnant women, children under 5 or more than 5000 participants you may need additional insurance cover. Insurance for studies, email the insurance team with any insurance enquiries.

If yes, please provide confirmation that insurance cover has been agreed.

- No

Q54. Has your research project been independently peer reviewed?

This can be organised by the Peer Review Office (within the RGIT). If you answered yes to any questions in section 3, you may be asked to ensure the study is peer reviewed. However, the study does not have to use the RGIT's office for peer review.

- No

Q55. Are you developing a mobile app?

See the mobile app webpage for more information.

- No

Q56. Have you had a Disclosure and Barring Service (DBS) check carried out? If yes, when (add date).
For more information about DBS, check government guidance and the College website.

- No

Q57. Do you need a contractual agreement in place? For further information, please contact your supervisor in the first instance.

- No

Q58. Do you have permissions to use the data in your study? This may be required if you are looking at secondary data.
Postings on social media may not be considered public for research uses – please check

- Public data, permission not required

Q59. Has Imperial College's Risk Assessment procedure been followed? Contact your departmental administrator for further information.

- Yes

Section 12: Confidentiality and management of personal and other research data

Q61. I understand it is the responsibility of the researcher to ensure all research data is securely stored during and after the study in accordance with College Guidelines, Codes of Practice, Policies and Procedures.

- Yes

Q62. I confirm that all the processing of personal information related to the study will be in full compliance with the GDPR. Including but not limited to, the creation of all necessary documentation (PIS, Data Protection Impact Assessments, Consent forms etc.)

- Yes

Q63. Describe how you will preserve the confidentiality of participants taking part in the study and fulfil transparency requirements under the General Data Protection Regulation.
Best practice requires that all participants information be entirely anonymised. If you are not doing this please give well-justified reasons for keeping personal and traceable information.
All data should be stored on a GDRP-compliant service e.g. Imperial College OneDrive or Sharepoint.

Not applicable

Section 13: Co-investigators/ Collaborators (second supervisor or external collaborators)

(to be completed by all)

If there are more than four co-investigators (second supervisor or external collaborators), please use a separate sheet and follow the format below.

Q66. Name of Co-investigator/ Collaborator

Nathan Johnson

Q67. Position Incl. organisation, company, institution

Research Associate in Sustainable Energy Systems, Imperial College London

Q68. Role in the study (what contributions you will make and relevant experience)

Second supervisor

Q69. Email (Work, not personal)

nathan.johnson17@imperial.ac.uk

Q70. Do you have another Co-investigator/ Collaborator (second supervisor or external collaborators)?

- No

PI Declaration

I declare that:

- I undertake to abide by the ethical principles underlying the Declaration of Helsinki (1964) and subsequent amendments and good practice guidelines on the proper conduct of research.
- I undertake to abide by the Data Protection Act 2018 and General Data Protection Regulation (Europe) and any applicable local laws.
- I undertake to abide by all local laws and regulations for non-UK research.
- I will report any adverse or unforeseen events or protocol violations and deviations which occur to the Ethics and Research Governance Co-ordinator within 24 hours.
- I will provide an annual progress report of the project until the end of the study.
- If I register my study on a public database, i.e. ClinicalTrials.gov, I will report results on that database within one year of study completion.
- I will provide notification of the end or early termination of the research project.
- I will provide notification of amendment to the CEP Ethics Panel if there are any changes to the research protocol or personnel which affect the ethical aspects of the project.
- I will assist the CEP Ethics Panel in any continuing review of the project deemed necessary by the Committee or Faculty Members.
- All information on this form is correct.

Q90. PI Name (please print)
Iain Staffell

Q91. PI Signature: PI must sign and date this section after you have submitted and downloaded the PDF form
(Note: PDF copy of form can be downloaded on final page of survey following submission)

N/A

Q92. Date
18 May 2026

Any attendance required by the CEP Ethics Panel must be made by the PI named in section one. Attendance at the meeting will give you the opportunity to answer any ethics questions raised by the committee.

Please note, this is the last section of the form. When you press the 'next' button, you will submit your ethics application form. Please ensure you are ready to submit, as you can't go back and amend once you press the 'next' button. Once submitted, all forms should be downloaded using the 'Download PDF' button on the final survey response page and then sent to your supervisor for checking and signing off.

Embedded Data:

N/A